

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456162.8332 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.041
Transit depth (Rp/Rs)^2	1.94 +/- 1.13 [%]
Orbital Inclination (inc)	89.61 +/- 1.43
Airmass coefficient 1 (a1)	1.15 +/- 0.33
Airmass coefficient 2 (a2)	-0.13 +/- 0.66
Scatter in the residuals of the lightcurve fit is	29.97 %
Best Comparison Star	#2 - [305, 109]
Optimal Aperture	2.76
Optimal Annulus	8.97
Transit Duration (day)	0.1245 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.139 $\pm$ 0.041 vs 0.1036 (deviation 0.83 σ)
Duration fit vs published	0.1245 d vs 0.1567 d (deviation 5.91 σ)
Mid-time O-C	8.9 min
Depth SNR	3.2
Residual scatter	29.97 %

Light curve

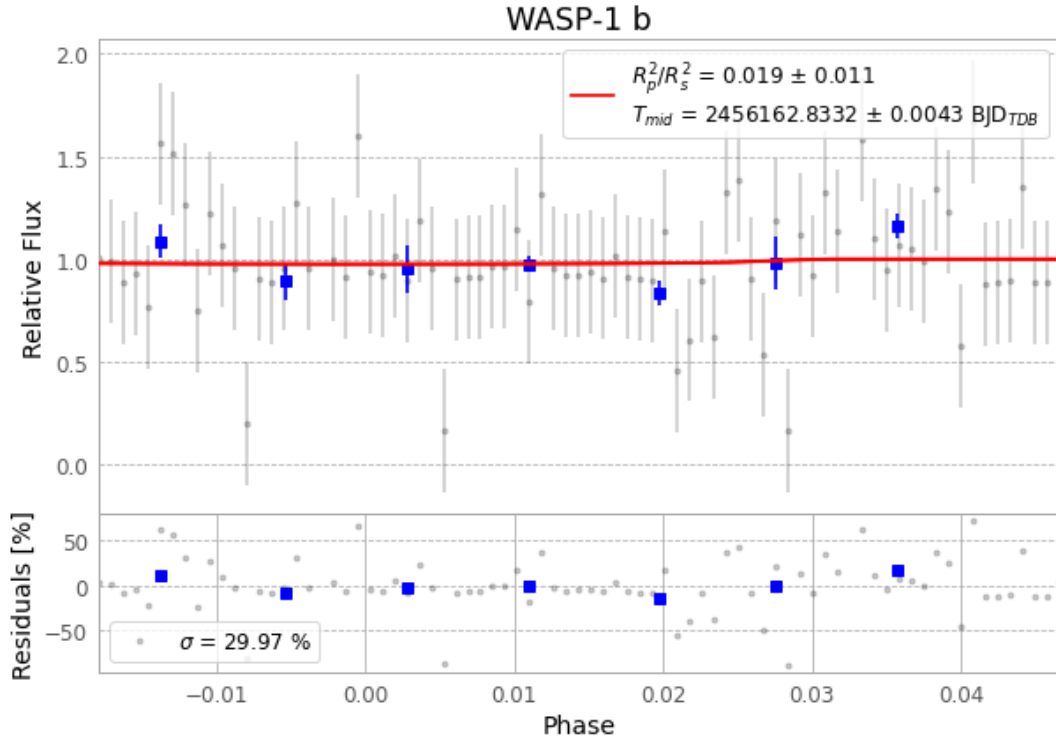


Figure 1 — FinalLightCurve_WASP-1 b_2012-08-22.png (SHA-256 c7b258fc4318430c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 224], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 3f81adf470ebee1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

79 FITS frames (52 MB), WASP-1120823064813.FITS → WASP-1120823104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-120823.log (e351eb25ecf5...), FinalLightCurve_WASP-1 b_2012-08-22.png (c7b258fc4318...), FinalLightCurve_WASP-1 b_2012-08-22.csv (ccaad4a83707...)

Compute resources

Wall time 169.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:21Z.